Lab09: Candy Mini Project

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Background

In today's mini-project we will analyze candy data with the exploratory graphics, basic statistics, correlation analysis and principal component analysis methods we have been learning thus far

Data Import

The data comes as a CSV file from 538.

```r
candy <- read.csv("candy-data.csv", row.names = 1)
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

Q1. How many different candy types are in this dataset?

There are 12 different candy types

Q2. How many fruity candy types are in the dataset?

There are 38 fruity candy types

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

My favorite candy Hershey's Kisses has win percent of 55.37545

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
candy |>
  filter(row.names(candy)=="Hershey's Kisses") |>
  select(winpercent)
```

```
                 winpercent
Hershey's Kisses   55.37545
```

Q4. What is the winpercent value for "Kit Kat"?

The win percent of Kit Kat has win percent of 76.7686

```
candy |>
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)
```

```
        winpercent
Kit Kat    76.7686
```

Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

The win percent of Tootsie Roll Snack Bars is 49.6535

```r
candy |>
  filter(row.names(candy)=="Tootsie Roll Snack Bars") |>
  select(winpercent)
```

```
                        winpercent
Tootsie Roll Snack Bars    49.6535
```

```r
library(skimr)
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The column `hist` looks different than any other, it looks like a graph

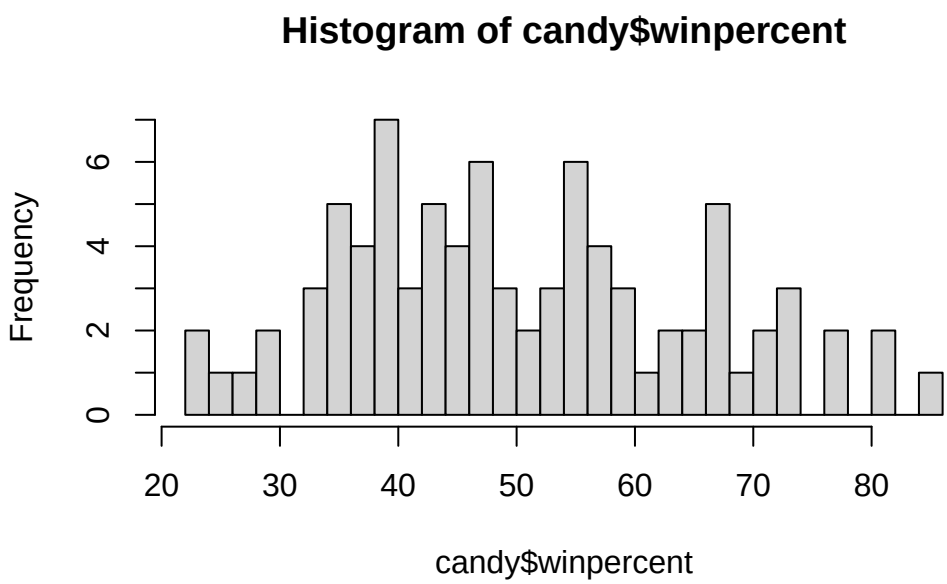
Q7. What do you think a zero and one represent for the candy$chocolate column?

It shows either a candy has chocolate in it or not, 1 means yes, 0 mean no.

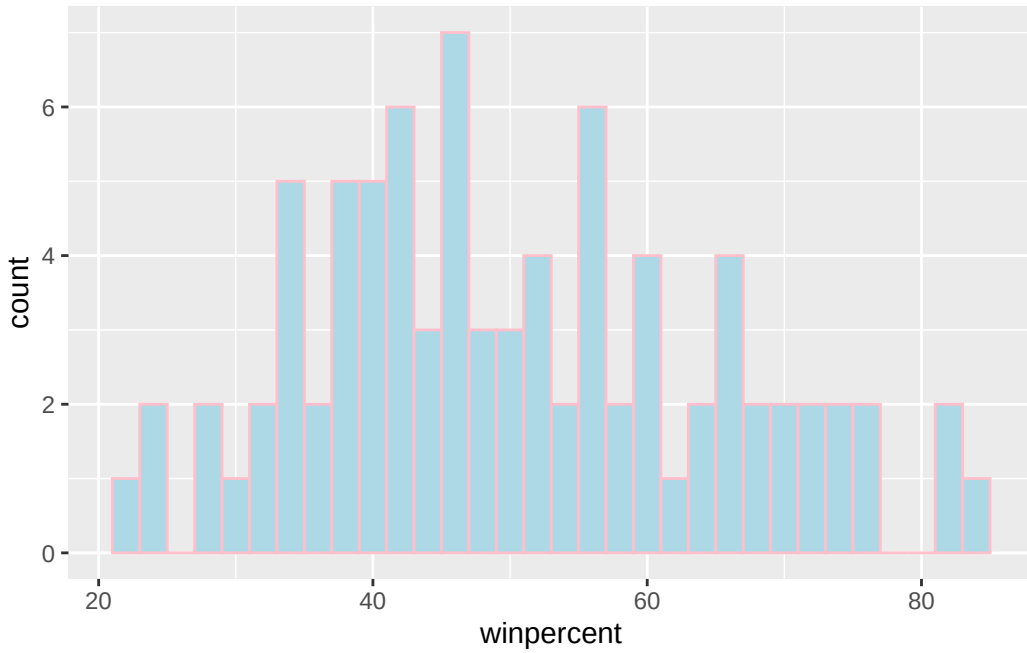
Exploratory Analysis

Q8. Plot a histogram of winpercent values using both base R an ggplot2.

```
library(ggplot2)
hist(candy$winpercent, breaks = 30)
```




```
ggplot(candy, aes(x = winpercent)) +
  geom_histogram(binwidth = 2, col = "pink", fill = "lightblue")
```



Q9. Is the distribution of winpercent values symmetrical?

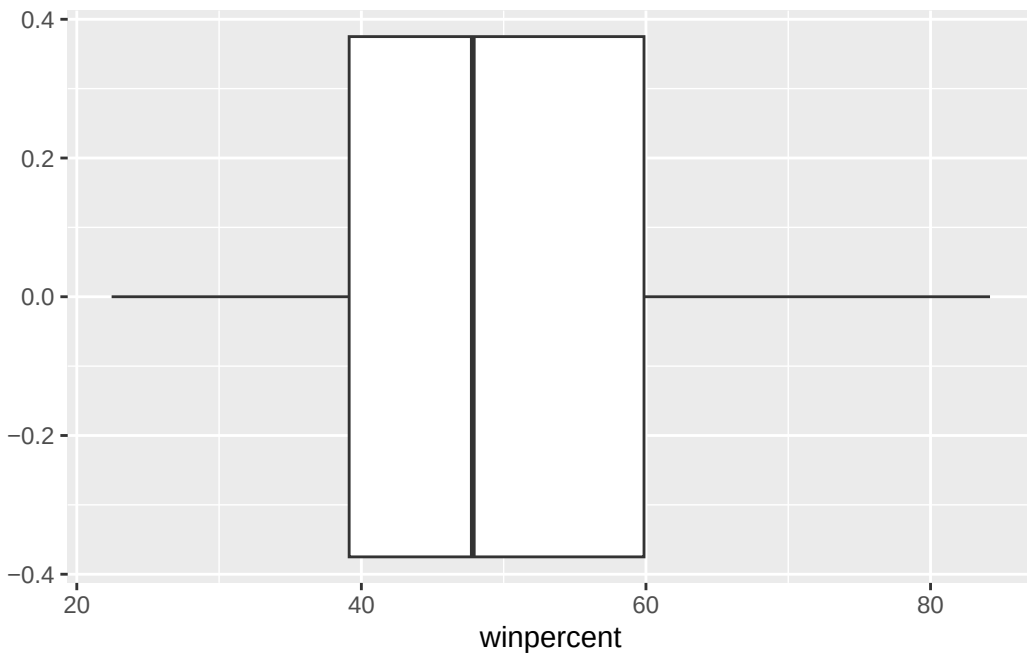
It doesn't look quite symmetrical

Q10. Is the center of the distribution above or below 50%?

```
summary(candy$winpercent)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  22.45   39.14   47.83   50.32   59.86   84.18 
```

```
ggplot(candy, aes(x=winpercent)) +
  geom_boxplot()
```



The center of distribution is above 50%

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
cho_candy <- candy$winpercent[candy$chocolate == 1]
cho <- mean(cho_candy)
fru_candy <- candy$winpercent[candy$fruity == 1]
fru <- mean(fru_candy)
print(paste("chocolate candy is" , ifelse(cho > fru, "higher", "lower") , "than fruit candy"
```

```
[1] "chocolate candy is higher than fruit candy"
```

Q12. Is this difference statistically significant?

```
t.test(cho_candy, fru_candy)
```

```
    Welch Two Sample t-test

data:  cho_candy and fru_candy
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
```

```
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

The difference is insignificant p-value is small

Overall Candy Ranking

Q13. What are the five least liked candy types in this set?

```r
inds <- order(candy$winpercent)
head(candy[inds, ], 5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
Boston Baked Beans                0    0   0        1        0.313        0.511
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
Jawbusters                        0    1   0        1        0.093        0.511
                   winpercent
Nik L Nip            22.44534
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

The 5 least liked candy types are "Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, Jawbusters".

Q14. What are the top 5 all time favorite candy types out of this set?

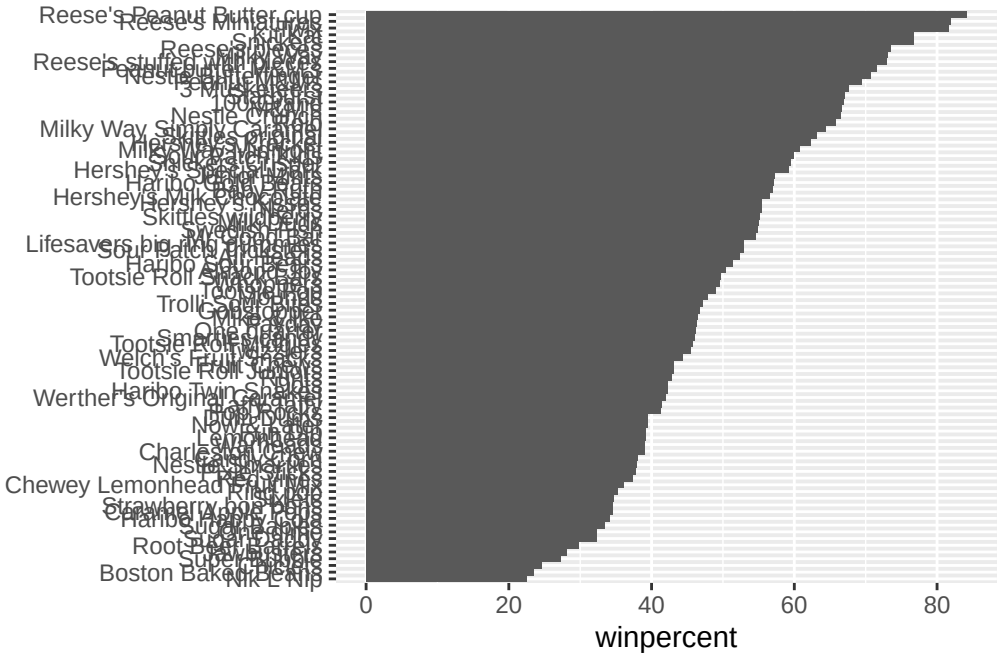
```r
inds <- order(candy$winpercent, decreasing = TRUE)
head(candy[inds, ], 5)
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup         1      0       0              1      0
Reese's Miniatures                1      0       0              1      0
Twix                              1      0       1              0      0
Kit Kat                           1      0       0              0      0
Snickers                          1      0       1              1      1
                          crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
Twix                                     1    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Snickers                                 0    0   1        0        0.546
                          pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

The 5 most favorite candies are "Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, Snickers"

inds <- order(candy$winpercent) head(candy[inds,], 5]

```r
ggplot(candy) +
  aes(winpercent,
      reorder(rownames(candy), winpercent) )+
  geom_col() +
  ylab("")
```




```
ggsave("barplot2.png", height=10, width=6)
```

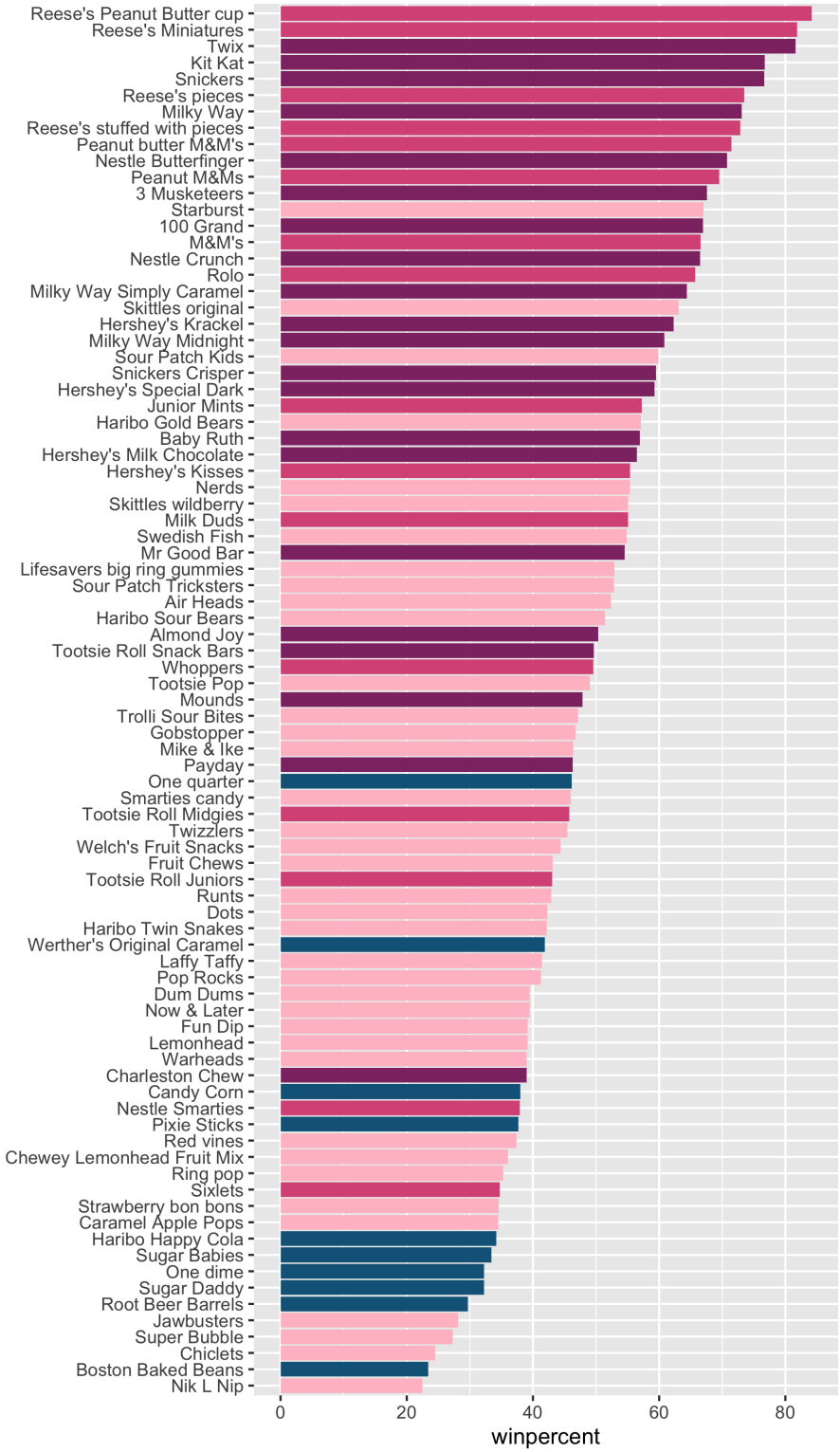
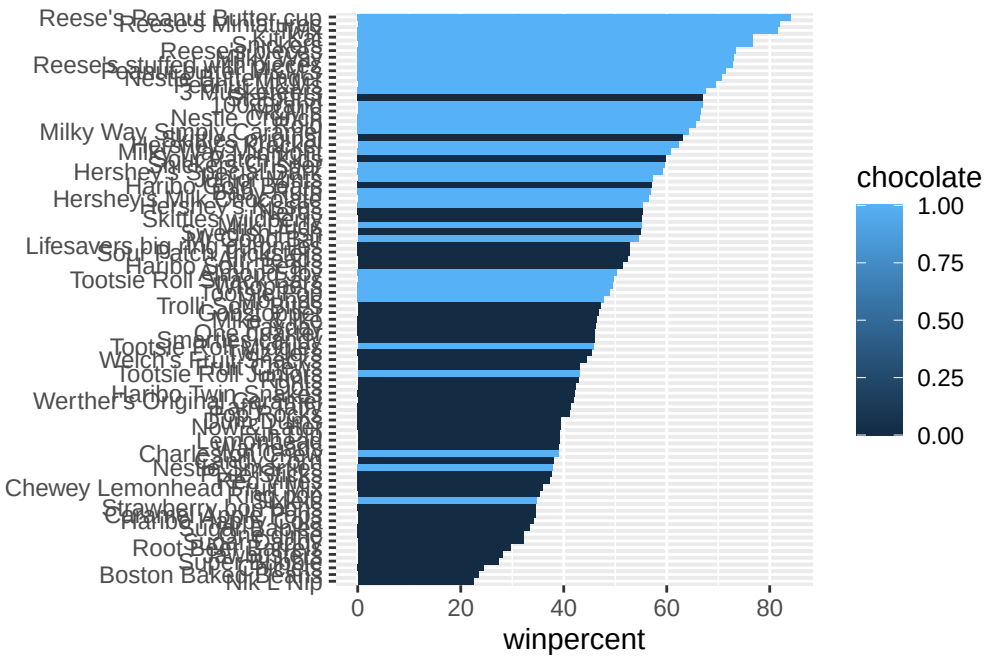


Figure 1: My second barplot for Q16

Time to add some useful color

Color by chocolate:

```
ggplot(candy) +
  aes(winpercent,
      reorder(rownames(candy), winpercent),
      fill=chocolate)+
  geom_col() +
  ylab("")
```



I want custom colors that I pick so we need to make this ourselves…

```
my_cols <- rep("#126388", nrow(candy))
my_cols[candy$chocolate==1] <- "#d95986"
my_cols[candy$bar==1]       <- "#903473"
my_cols[candy$fruity==1]    <- "pink"
ggplot(candy) +
  aes(winpercent,
      reorder(rownames(candy), winpercent))+
  geom_col(fill=my_cols) +
  ylab("")
```

```
ggsave("barplot2.png", height=10, width=6)
```

Q17. What is the worst ranked chocolate candy?

```
worst <- order(candy$chocolate)
worst_name <- candy[worst, ][1, ]
worst_name
```

```
         chocolate fruity caramel peanutyalmondy nougat crispedricewafer hard
One dime         0      0       0               0      0                 0    0
         bar pluribus sugarpercent pricepercent winpercent
One dime   0        0        0.011        0.116   32.26109
```

```
best <- order(candy$fruity)
best_name <- candy[best, ][1, ]
```

The worst ranked chocolate candy is sixlets

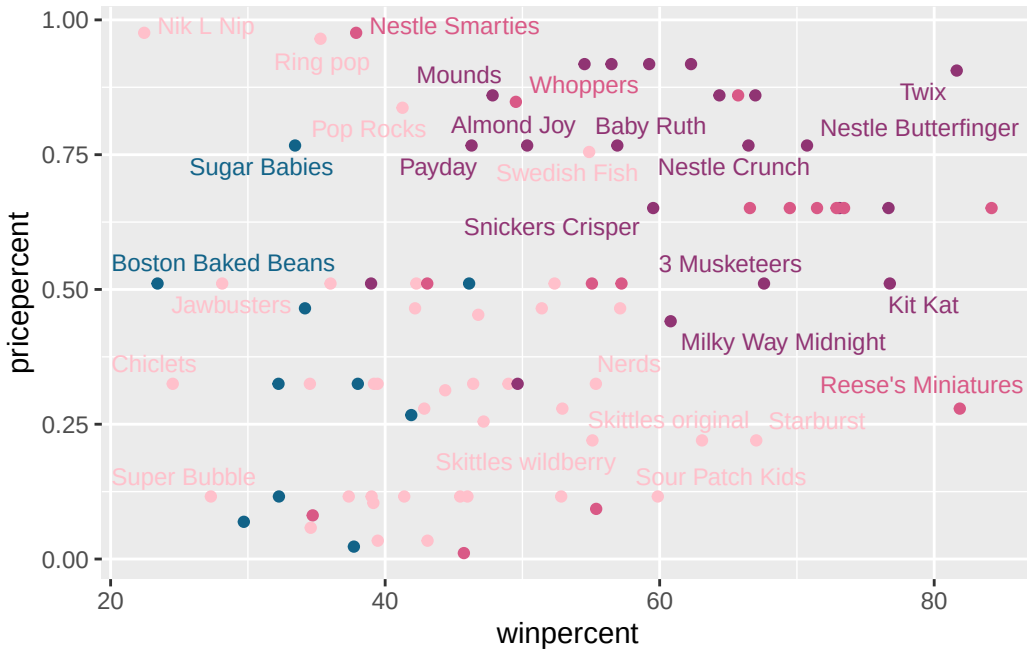
Q18. What is the best ranked fruity candy?

The best ranked fruity candy is starbust

```r
library(ggrepel)

ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 7)
```

```
Warning: ggrepel: 57 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Miniatures

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```r
inds <- order(candy$pricepercent, decreasing = TRUE)
head(candy[inds, ], 5)
```

```
                         chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                        0      1       0              0      0
Nestle Smarties                  1      0       0              0      0
Ring pop                         0      1       0              0      0
Hershey's Krackel                1      0       0              0      0
Hershey's Milk Chocolate         1      0       0              0      0
                         crispedricewafer hard bar pluribus sugarpercent
Nik L Nip                                0    0   0        1        0.197
Nestle Smarties                          0    0   0        1        0.267
Ring pop                                 0    1   0        0        0.732
Hershey's Krackel                        1    0   1        0        0.430
Hershey's Milk Chocolate                 0    0   1        0        0.430
                         pricepercent winpercent
Nik L Nip                       0.976   22.44534
Nestle Smarties                 0.976   37.88719
Ring pop                        0.965   35.29076
Hershey's Krackel               0.918   62.28448
Hershey's Milk Chocolate        0.918   56.49050
```

The 5 most expensive are "Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, Hershey's Milk Chocolate" and lest popular is Nik L Nip

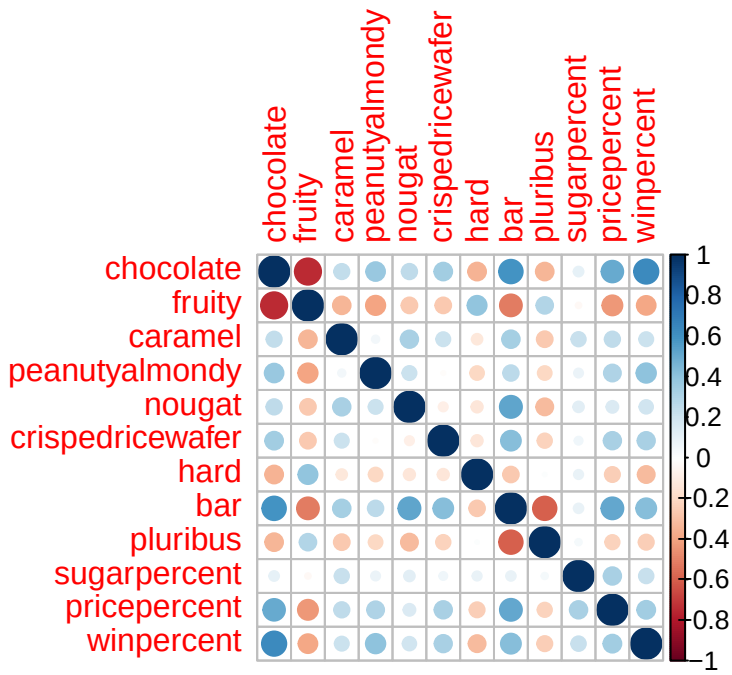
Exploring the correlation strucutre

Pearsonn correlation values range fram -1 to +1

```r
library(corrplot)
```

```
corrplot 0.95 loaded
```

```r
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The fruity annd chocolate

Q23. Similarly, what two variables are most positively correlated?

The bar and chocolate also winpercent

Principal Component Analysis

```
pca <- prcomp(candy, scale=T)
summary(pca)
```

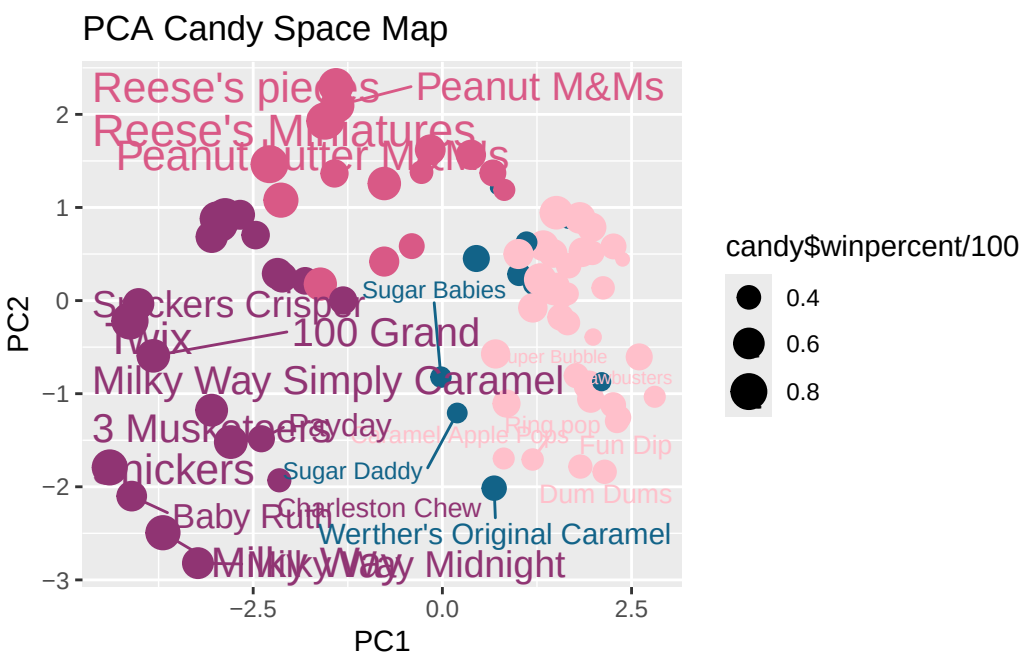
```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
```

```
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

The main results figure: the PCA score plot:

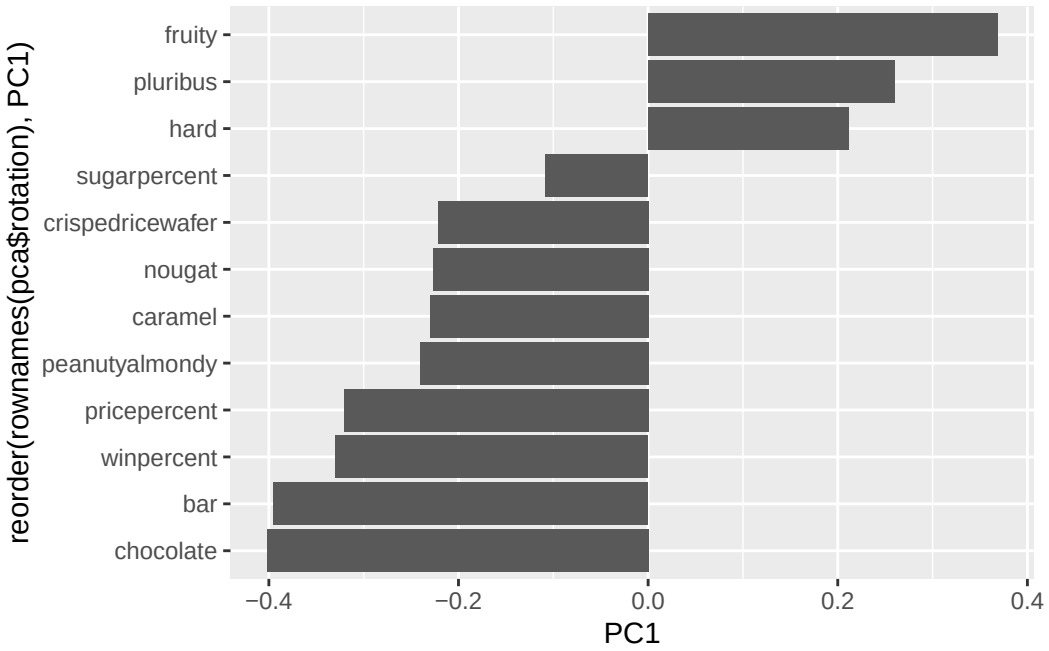
```
p <- ggplot(pca$x) +
  aes(PC1, PC2, size=candy$winpercent/100, label=rownames(pca$x)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols) +
  labs(title="PCA Candy Space Map")
p
```

```
Warning: ggrepel: 61 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



The "loading" plot for PC1

```
ggplot(pca$rotation) +
  aes(x = PC1, y = reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



> Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

This makes sense, the values are bar, chocolate, winpercentage and pricepercentage

```
#library(plotly)
#ggplotly(p)
```

> Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A winning candy should be a bar chocolate. The price does not negativelly affect the win percentage, showing good business. The PCA supports the correlation of most chocolate are closely correlation with wins. Visualization shows clear data, and bigger circle with higher win percentage